Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/05/2023
PI Name:	Pecaut, Michael Ph.D.		
Project Title:	Impact of Spaceflight on Primary and Secondary Antibody Responses		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	mpecaut@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8372
Organization Name:	Loma Linda University		
PI Address 1:	Division of Biomedical Engineering Sciences, Department of Basic Sciences		
PI Address 2:	11175 Campus St, Chan Shun Pavilion, A-1016		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	07/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/31/2023 per NSSC information (Ed., 6/20/22) NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 10/14/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/21) NOTE: End date changed to 3/30/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 09/30/2020 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 11/30/2018 per NSSC (Ed., 10/10/17) NOTE: End date changed to 11/30/2017 per NSSC (Ed., 6/3/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chapes, Stephen Ph.D. (Retired (formerly from Kansas State University))		
Grant/Contract No.:	NNX13AN34G		
Performance Goal No.:			

Performance Goal Text:

The spaceflight environment influences immune cell distribution, ex vivo responses to antigen, and cytokine expression. However, few studies have characterized the impact of the spaceflight environment on the host immune response following an in vivo immune challenge, particularly those involving B cells and antibody production. The International Space Station (ISS) provides an ideal laboratory within which investigators can explore this unique environment and its impact on the host response. Through the use of a mouse model (with an immune system that closely parallels that of humans), we will establish a link between exposure to the spaceflight environment and decrements in immune system activation. This study is significant because it will directly test the hypothesis that host immunological health will be compromised in space.

Task Description:

 On-orbit Flight Activities, Procedures, and Timeline

Mice will receive health checks upon unloading and transfer from the Transporter to the Habitat after capsule docking no later than 24 hrs after hatch opening. Animal health will then be monitored daily via continuous video recording from one camera at a time from all of the Habitats on-orbit (including other investigations which may be running concurrently) and down-linked to Ames science staff. Ground personnel shall have the option of programming or manually controlling which camera is being down-linked and recorded at any given time. Water levels will be checked every 3-4 days. After a minimum of 14 days after animal transfer from the Transporter to the Habitat, all 20 mice will be moved to the MSG and receive intraperitoneal (i.p.) injection of vehicle (5 mice), Tetanus Toxoid (TT) vaccine (5 mice), CpG ODN adjuvant (5 mice), or TT vaccine + CpG ODN adjuvant (5 mice). The dose of TT and/or CpG ODN will be 10 microg/ mouse and 40 microg/ mouse, respectively. Pre-mixed doses will be prepared preflight in 100 microl saline and stored at -20 to -30°C until the appropriate time point. All injections and blood draws will use a 1 mL 26G 3/8” tuberculin syringe. All injections will be i.p.

Before use, the CpG adjuvant must be stored at -20°C or colder until use and the TT vaccine must be stored at 4°C. At 14 +/- 2 days after injection, mice will be transferred to the MSG and sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Following spleen collection, the remains of the mouse carcass will be wrapped in tin foil, put in a zip-lock bags and placed in a -80°C environment (or colder) to preserve the tissues for future biospecimen sharing. Blood must not be allowed to sit at ambient temperature for longer than 1 hour post-collection; if operational requirements require the blood collected to be stored in excess of 1 hour before centrifugation, the blood must be transferred to a 4°C environment until centrifugation, for a period not to exceed 4 hours. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to Earth. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.
 <table border=1 cellspacing=0 cellpadding=0 align=center width=672> <tr > <td width=91> Time </td> <td width=252> Activity & Estimated Time Required </td> <td width=76 > Thermal Req. </td> <td width=253 > Comments, Location & Special Considerations </td> </tr> <tr > <td width=91 > Launch to L + 45 days </td> <td width=252 > Animal health checks via video monitoring; water level checks every 3-4 days </td> <td width=76 > 20 – 30°C </td> <td width=253 > Video downlink required for health checks </td> </tr> <tr> <td width=91 > Transfer to Habitat + 14 d </td> <td width=252 > All 20 mice will be transferred to habitat </td> <td width=76 > 20 – 30°C </td> <td width=253 > Requires transfer to and from Habitat with MTB; </td> </tr> <tr > <td width=91 > Injection + 14d +/- 2 d </td> <td width=252 > At specific time points, all 20 mice will be removed from AEM and given appropriate injection </td> <td width=76 > 20 – 30°C / -80°C </td> <td width=253 > Required transfer to MSG for injections </td> </tr> <tr > <td width=91 > Dissection + 0 d </td> <td width=252 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen. Blood and spleen samples kept at -80^oC until delivered to laboratory of PI </td> <td width=76 > -80°C </td> <td width=253 > Required transfer to MSG for dissection, processing of blood in the centrifuge, and storage of samples at -80°C for up to 6 months. </td> </tr> </table>

  Hardware Housekeeping Operations: Crew-performed

Daily hardware checks will occur. Habitats will be kept at a regular 12hr:12hr day:night light cycle to match the day/night programmed to be synchronous with the crew’s day/night cycle upon arrival. Water refill of habitats shall be performed. To prevent rodent food bars from becoming too hard or soiled, they will be replaced with a fresh food bar plate (from 4°C storage) at a time to not exceed 30 days after the introduction of mice to a Habitat.

Ground Activities Accompanying On-orbit Activities and Timeline

Ground controls will be run following spaceflight environmental parameters in orbital environmental simulator(s) (OES) chambers at KSC ACF. Delay between launch and start of environmental controls may be up to 8 days, depending on the logistics and capability of the telemetry system. Mice will be housed in flight Habitats with RFB diet and lixit water to match on-orbit groups. The operations done on ground will be conducted to match the same relative time points during the day/night cycle of the animals on-orbit, so that procedures on ground control mice are conducted at the same points in the animals’ circadian rhythm.

Each experimental group for ground controls will be doubled to improve the statistical significance of the experiment with a total of 40 mice (4 groups of 10). Therefore, 2 Transporters will be required for the flight in orbital environmental simulator(s) (OES) that have been programmed with telemetry data from the flight. At the same relative time point (accounting for delay in starting ground controls after flight) the mice are transferred from the Transporter to the Habitat, the 40 ground control mice will be transferred from the 2 Transporters into 4 Habitats, and maintained in OES programmed with telemetry data from the ISS. OES shall be programmed to mimic the following conditions experienced by the Transport and then the Habitat: CO2, O2, temperature, and humidity.

Injection and dissection must be performed as close as possible to how they are done on-orbit. In particular, the time of

day that mice are sacrificed and dissected on the ground should match the same point in the circadian rhythms of the mice on-orbit. However, dissections or ground control mice can be conducted on standard laboratory benches/tables with chairs that are more ergonomic to the operators. For ground controls, dissection pins or needles may be used to pin down mice for dissection, as can spray bottles of wetting the mice before dissection. Flight hardware is required when it may affect the tissue dissection or preservation, such as having flight cold blocks in mini cold bags for tissue freezing.

Mice will receive health checks daily via video in the Habitat and water levels will be checked every 3-4 days. After a minimum of 14 days in the OES, all 40 mice will receive an i.p. injection of vehicle (10 mice), TT vaccine (10 mice), CpG adjuvant (10 mice), or TT vaccine + CpG adjuvant (10 mice). At 14 +/- 2 days after injection, mice will be sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Duration for which blood is stored at ambient or 4°C shall be matched as closely as possible to the timelines the same procedures will be performed on-orbit. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to the laboratory of the PI. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.

<table class=MsoNormalTable border=1 cellspacing=0 cellpadding=0 align=center width=672 > <tr > <td width=145 > Time </td> <td width=186 > Activity </td> <td width=66 > Thermal Req. </td> <td width=275 > Comments, Justification, Special Considerations </td> </tr> <tr > <td width=145 > Beginning of ground controls to +45 days </td> <td width=186 > Animal health checks; water level checks every 3-4 days </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Video downlink required for health checks </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > At specific time points, all 40 mice will be removed from AEM and given appropriate injection </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Transfer from cages to a regular lab bench and back w/ standard cages. </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen </td> <td width=66 > -80^oC </td> <td width=275 > Processing of blood in a centrifuge, and storage of samples at -80^oC </td> </tr> </table>

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

According to the National Institute of Aging (NIA), “Life expectancy nearly doubled during the 20th century with a ten-fold increase in the number of Americans age 65 or older.” Furthermore, “there are approximately 35 million Americans age 65 or older, and this number is expected to double in the next 25 years.” Consequently, there is a rapidly growing need to understand the mechanisms underlying age-associated illness. Indeed, in the last decade, research has repeatedly demonstrated an impact of aging on immune function. Low-grade increases in circulating inflammatory cytokine secretion are often associated with senescence. Furthermore, studies suggest that these disruptions are likely due to decrements in cellular repair mechanisms and/or increases in antigenic load. Given this disruption in immune function, it is perhaps not surprising that many of the diseases associated with aging — such as type 2 diabetes, rheumatoid arthritis, Parkinson’s disease, Alzheimer’s disease, cardiovascular disease — have strong inflammatory underpinnings.
There are strong indications that the physiological effects of spaceflight are also similar to those seen in senescence. For instance, evidence suggests that age-associated enhancements in inflammatory cytokines are linked to changes in adipose tissue and energy balance. Similar links between metabolism, adipose tissue, and inflammation have also been shown to manifest in astronauts and animal models after exposure to spaceflight. We propose to use long-term exposure to the spaceflight environment to model the effects of aging on immune function. Although these initial studies involve mice, the next obvious step in this work will be to work directly with astronauts. Because astronauts tend to be extremely healthy, these effects will not be influenced by confounding factors such as clinically relevant, age-related disease. Also, because the effects of spaceflight on immunity are temporary, generally returning to baseline levels post-flight, this will ultimately give us some insight into whether or not the effects of aging are, indeed, reversible.

Although many different aspects of immune function are influenced by both aging and spaceflight, these experiments will focus on B cells and the generation of antibodies. Specifically, we will use next generation RNA sequencing techniques to characterize the B cell Ig heavy chain in animals that are challenged with tetanus toxoid. The Ig heavy chain is the component of antibodies responsible for identifying specific sections of the tetanus toxoid, thereby coordinating subsequent immune responses. Comparisons will be made between the B cell repertoires generated on ISS and on Earth. The changes in B cell Ig heavy chain usage will provide indications whether the spaceflight environment affects the B cells that respond to antigen. This, in turn, will help us understand the impact of aging on B cell function.

We anticipate that we will obtain several data sets which have relevance to spaceflight and/or have general significance to understanding host immune responsiveness. These data include: 1) Data that show that spaceflight either maintains or alters the basic distribution of B cells based on Igh transcripts; 2) Information that demonstrates how immunization with an immunogen, commonly used in human medicine, alters the basic distribution of B cells based on Igh transcripts in the absence of the spaceflight variable; 3) Determinations how activation of B cells through Tlr9 receptors alters the basic distribution of B cells based on Igh transcripts.

We will ascertain if activation of B cells through Tlr9 receptors can reverse any impact that may occur during spaceflight.

Project History and Review of Specific Aims
This project was submitted and funded in response to the NASA Research Announcement NNH12ZTT001N: Research Opportunities in Space Biology. This multidisciplinary project was to include three independent investigators and two institutions (Loma Linda University/LLU and Kansas State University/KSU, in addition to NASA). Since then, due to a series of unpredictable events, this project has evolved to include a number of ground-based analog studies, multiple analyses of flight tissues, and an actual flight study. We designated this experiment as Tetanus Antibody Response from B-cells In Space (TARBIS).

The original project involved three different experiments aboard the International Space Station (ISS) to characterize different aspects of the antibody response. We proposed to immunize mice with the established tetanus toxoid (TT) vaccine. This would allow us to generate an antigen-specific antibody response in our mouse model while using a clinically-relevant antigen. Our goal was to determine the impact of the spaceflight environment on primary and secondary antibody responses. C57BL/6J mice were challenged with a TT vaccine either before launch or during their stay on the ISS, as appropriate. Blood and spleen were to be collected in orbit from mice 2 weeks after primary or secondary immunization. Vaccination and assessment schedules would have been matched in ground-based controls. The antibody response and the B cell immunoglobulin heavy-chain (IgH) repertoire would be determined in the splenic B cells from each treatment groups and compared among flight and ground treatment groups.

Due to the loss of astronaut work time assigned to Space Biology and a greater-than 2-year delay in the transport of mice to the International Space Station, we were approved in 2015 to modify the aims of NNX13AN34G to include a Flight Definition Phase. Much of the work between 2015 and 2018 involved completing the Flight Definition phase work and publishing the results. In 2017-2018, we requested a funding augmentation and began preparations for our spaceflight mission. As this process evolved, two of our originally proposed experiments were combined into one flight experiment. During this period a Co-Investigator, Dr. Keith Chapes, announced that he would be retiring from KSU. With his retirement, all experimental operations were moved to LLU. Previously, this work was completed at KSU in Dr. Chapes' lab. Dr. Chapes continued his role as a Co-I as an independent consultant. In 2019, this experiment flew as TARBIS/Rodent Research-12. In the years between 2019 and the present, we were involved with a number of other flight-related opportunities, including several Biospecimen Sharing Program dissections and RR-18. Finally, the COVID-19 pandemic significantly disrupted and delayed multiple activities during this period. Flight Definition Phase Work

1. Analysis of spleen tissue/RNA from the mice flown on Rodent Research - 1 (RR-1) verification flight.

We analyzed spleen RNA from these tissues (10 mice; 5 flight and 5 ground controls). We have determined IgH usage of the spleen B cells in these mice (flight vs. ground controls). We also generated total RNASeq on the RNA to have a matching data set to the liver RNASeq that is being done by the NASA GeneLab.

We analyzed GeneLab sequencing data that was available from the GeneLab database (GLDS-47, GLDS-48). The number of immunoglobulin reads (either heavy or light chain) were quite low, less than 1% of what we had obtained in preliminary analysis of spleen cells and tissue. We first attributed this low yield to short sequencing reads from the HiSeq sequencing method that Genelab did and provided in the database. Therefore, we sequenced both splenic (funded by another grant) and liver RNA from the Rodent Research-1 (RR-1) validation flight, which was made available by the NASA Ames Research Center (n=3 ground, n=3 flight). RNA was sequenced on the Illumina MiSeq platform (300bp, 2x) at the Kansas State Integrated Genomics Facility. Sequencing data was quality trimmed using CLC Genomics Workbench and mapped to reference libraries obtained from the ImMunoGeneTics (IMGT) information system and the National Center for Biotechnology Information (NCBI) gene database. Data sets were submitted to the IMGT HighV-Quest tool for Complementarity Determining Region 3 (CDR3) characterization.

The analysis of liver (quantified as part of the tissue sharing grant, NNX15AB45G) vs. spleen (quantified as part of THIS grant) reads indicated that liver immunoglobulin gene reads range from 0.26% of spleen immunoglobulin gene reads to 0.91%. The number of reads ranges from less than 100 to no greater than 622, depending on the sample. Although useful, publishable data from the spleen can be obtained (preliminary data presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) and presented in the GeneLab workshop by Chapes and Ward), the liver sequencing will not provide meaningful immunoglobulin gene data, even when the more comprehensive MiSeq next generation sequencing (NGS) platform is used to sequence. The liver data won't be of any value and will not fulfill our goal of doing tissue comparisons. Other data from the MiSeq NGS data sets will be useful for other data mining attempts and will be made available to the GeneLab. Therefore, these sequencing runs allowed us to narrow our focus and not waste time on liver analyses. [Ed. Note: For information on the tissue sharing grant, see NASA Grant # NNX15AB45G; PI: Chapes. For ASGSR and other References, see the Cumulative Bibliography for this project.]

2. An antiorthostatic suspension (AOS) done to skeletally unload mice to simulate the skeletal unloading that occurs during spaceflight.

In this protocol, mice were subjected to AOS. Two weeks later the mice were immunized in 1 of 4 ways: phosphate buffer saline (PBS), unmethylated cytosine–guanine dinucleotide motifs (CpG), tetanus toxoid (TT), and TT with CpG. Two weeks after immunization and continued AOS, we sacrificed the mice and collected a variety of tissues for further analysis. These are the same treatment groups that were proposed for our ISS studies. We used female C57BL/6J mice, 10-12 weeks old at the start of unloading. Independent variables were Tetanus Toxoid, CpG adjuvant, and hindlimb unloading. Blood and spleen were analyzed as part of this grant (NNX13AN34G). Assays at Loma Linda University included blood, TT specific IgG, IgA, IgM, and non-specific Ig levels. Kansas State University completed analyzing the IgH & IgK repertoire based on RNAseq.

The AOS was done to skeletally unload mice to simulate the skeletal unloading and stress that occurs during spaceflight (Chapes, Mastro et al. 1993). [Ed. Note: See References.] Experiment 1 took place in March 2016 (5 mice per treatment group); Experiment 2 took place in May 2016 (5 mice per treatment group). In this protocol, female, 8-week-old mice were subjected to AOS. Two weeks later, the mice were immunized in 1 of 4 ways: PBS (buffer control), CpG only (adjuvant), tetanus toxoid (TT; antigen) only, and TT with CpG. Injections were done by intraperitoneal (i.p.) route. Two weeks after immunization and continued AOS (28 day total unloading), we sacrificed the mice and collected a variety of tissues. This protocol was followed so that when mice were flown to the ISS, we would have the AOS as a comparison group with parallel variables. Serum, spleen, liver, and bone marrow were collected specifically to compare with our flight project.

Briefly, we were able to confirm that i.p. injections – as opposed to the more typical intramuscular (i.m.) injection for vaccines – of TT resulted in an antibody response and that this response was augmented by an order of magnitude when mice were also given the CpG adjuvant. However, AOS had little impact on the ability to generate TT-specific antibodies, at least in the blood. AOS increased circulating corticosterone levels and a number of cytokines/chemokines, including GCSF, IL-1a, IL-5, IL-6, IP-10, and KC.

Impact of AOS on the Spleen Ig Repertoire

We explored the impact of skeletal unloading using the AOS model and mice were experimentally challenged with TT and/or the adjuvant CpG. AOS did alter constant gene usage, resulting in higher levels of IgM and lower levels of IgG. CpG also altered constant gene region usage, increasing usage of IgA. Significant changes could be detected in multiple V-, D-, and J-gene segments in both the heavy and light chains in response to AOS, TT, and CpG treatments. Analysis of class-switched-only transcripts revealed a different pattern of V-gene segment usage than detected in the whole repertoire and also showed significant alterations in gene segment usage after challenge. Alterations in V/J pairing were also detected in response to challenge. CDR3 amino acid sequence overlaps were similar among treatment groups, though the addition of CpG lowered overlap in the heavy chain. We isolated 3,045 whole repertoire and 98 potentially TT-specific CDR3 sequences for the heavy chain and 569 for the light chain. Our results demonstrate that AOS alters the repertoire response to challenge with TT and/or CpG.

Impact of AOS on the Bone Marrow Ig Repertoire

In the same AOS experiment, the bone marrow antibody repertoire was characterized – including V-, D- (heavy only), and J-gene segment usage, constant region usage, CDR3 length, and V(D)J combinations. We detected changes in gene-segment usage in response to AOS, TT, and CpG treatment in both the heavy and light chains of bone marrow B cells. Additionally, changes were seen in the class-switched VH-gene repertoire. Alterations were also detected in V/J pairing for both the heavy and light chains, and changes CDR3 length. We also detected lower levels of CDR3 AA overlap than detected in the splenic repertoire. These results demonstrate that AOS, TT, and CpG alter the bone marrow antibody repertoire; however, it is still unclear from both the spleen and the bone marrow data whether there is a loss of host antigen-specific responsiveness because of the change in gene use.

MiSeq analysis of the spleen was completed at KSU. Originally, we had intended to perform a similar analysis of the liver. However, as described above, this analysis was ineffective and we opted to analyze bone marrow instead. In addition, we characterized changes in circulating corticosterone and found that AOS led to a significant increase. Furthermore, we noted a trend for the AOS response to be augmented in TT treated mice.

Although the tissue analyses were slightly different from those described in the original proposal, the aim of doing multiple tissue assessments/comparisons was accomplished. This is a high-science return. We also believe that results from these studies will augment the data from the flight experiments, confirming the efficacy of AOS as a model for spaceflight for immune endpoints.

3. NASA Ames Research Center (ARC) Anti-orthostatic Suspension Experiment in Collaboration with Dr. Elizabeth Blaber.

Due to the nature of the studies described above, we were unable to use the blood or spleen tissue collected in the LLU experiments described above to characterize the impact of unloading on WBC populations. Furthermore, the funding and flight requirements for our flight project did not allow us to compare male vs. female mice. Fortuitously, the opportunity arose to look at blood populations in a study with a design similar to our own. This project involved Elizabeth Blaber, Ph.D. at Ames Research Center. [Ed. Note: See Space Biology projects under PI: Blaber in the NASA Task Book.]

The focus of Dr. Blaber's study was to study the impact of p21 on bone, muscle, and stem cells. Male and female mice were suspended for 15 or 30 days (the latter timepoint is very similar to the 28 day unloading in our AOS studies). At the appropriate timepoint, mice were sacrificed and tissues were collected immediately. We characterized blood populations using an ABC Vet automatic hematology analyzer. In addition, we collected adrenals, brain, cerebellum, eye, liver, and skin tissue and froze them in liquid nitrogen. Although we were primarily interested in the wild type controls for this study, we collected tissues from both wild type controls and the p21 knockout mice.

Task Progress:

Our analyses of the blood indicated that unloading decreased all of the major leukocyte population counts. This decrease was generally more pronounced in the lymphocyte subset than in the other populations. Female mice tended to have higher cell counts than males, but there was no significant unloading x sex interaction on any subset. Finally, counts were similar between wild-type controls and p21 knockouts for all populations.

4. Tetanus Toxoid Validation Test

In 2018, we were told that our flight experiment would be the focus of Rodent Research - 12 (RR-12). Because our initial assessment of the tetanus toxoid was several years old, we ran a small verification test on a new lot of the toxoid to ensure that biological activity of the vaccine would be as expected during the launch. This dose response study focused around the dose optimized in previous tests. Endpoints include TT-specific IgG levels. This experiment was performed in March of 2019. While we were able to verify that the same dose of TT could be use with in RR-12, we found that the CpG adjuvant was not as effective as in previous experiments. As a consequence, we were approved to double the CpG dose as part of the flight experiment.

5. An Analysis of the Effects of Spaceflight and Vaccination on Antibody Repertoire Diversity

As part of this project, our group had the opportunity to examine the antibody repertories in C57Bl/6 mice after a number of different situations. These data allowed us a unique opportunity to test the hypothesis that antibody diversity can be altered by physiological adaptation. That is, spaceflight, simulated spaceflight and/or immune responses are disruptions in stable environment and affect antibody diversity. To test this hypothesis, we analyzed antibody diversity, as expressed by CDR3 amino acids, in four experiments and five data sets we have generated, using analyses frequently used to assess communities in ecological contexts; specifically, the Shannon Index (SI), an established measure of diversity. We discovered a trend towards lower diversity as a result of spaceflight but did not find the same decrease in our physiological model of microgravity in either the spleen or bone marrow. However, the bone marrow repertoire showed a reduction in diversity after vaccination. We also detected differences in SI between experiments and tissues. We did not detect a pattern of CDR3 usage across the experiments. Overall, we were able to find differences in the

antibody repertoire diversity across experimental groups and tissues.

6. A Comparison of Unamplified and Massively Multiplexed Polymerase Chain Reaction (PCR) Amplification for Murine Antibody Repertoire Sequencing

We started this project with the hypothesis that an unamplified Immunoglobulin gene data set would provide comparable results to those seen in the commercially amplified data sets. However, there were no immunoglobulin repertoire studies done to compare data obtained using amplification techniques compared to the repertoire in a total RNASeq library. Concurrent with performing these analyses, we discovered that the required sample preparation for commercial sequencing also varied. Therefore, we found it necessary to examine the impact of sample preparation as part of our effort. This preflight verification work compared a data set generated using unamplified total RNA (TRNA) to commercially amplified data sets. We examined the role of commercial amplification and cDNA generation methods, as well as the impact of the starting material on sequence output. We found that sequencing of unamplified splenic RNA provides a realistic snapshot of the total splenic B cell repertoire. We also demonstrated that a good understanding of the bioinformatics workflow and reporting of the methodology is critical and cannot be understated. We established that there are cellular biases and transcript stability differences within B cell subpopulations. However, for the purpose of assessing a whole tissue B cell repertoire, unamplified RNASeq was capable of providing a valuable glimpse of the most prevalent B cell clones. Most importantly, an unamplified data set may detect V-gene segments that amplified data sets miss.

Flight Activities

Our flight investigation was launched on NG-11 out of NASA's Wallops Flight Facility (Wallops) on 4/17/19. At the request of NASA, this flight experiment combined experiments 1 and 2 of our original proposal, including: 1) a primary challenge in space, and 2) a primary challenge pre-flight + a secondary challenge in space. Members of our team participated in the baseline, environmental control, and vivarium control vaccinations and tissue collection at NASA Kennedy Space Center (KSC). All of these events were on a 3-4 day delay after similar procedures were performed by the ISS astronauts. Collected tissues for ground controls were frozen and shipped to LLU and stored in our freezers at -80°C. A subset of the flight blood and spleen samples were returned from the ISS and shipped to LLU to our lab in August 2019. The remaining samples and frozen carcasses of our flight mice were returned and shipped to LLU in January 2020.

We performed a tetanus toxoid specific IgG/IgA/IgM enzyme-linked immunosorbent assay (ELISA) on the blood samples from this study. There were some issues with the samples brought back from the ISS. Specifically, there was significant lysing of the red blood cells (RBCs) and we were unable to separate the cells from the plasma in roughly half of the cells. However, despite these difficulties, we were able to detect Ig in the appropriate samples.

The data suggest that both the CpG adjuvant and a secondary inoculation with the vaccine increased the antibody responses as expected on the ground. The difference in scale for the primary vs. secondary response figures indicates that under normal circumstances, B cell memory developed and responded as expected. However, both the primary and secondary responses were drastically inhibited after flight. There are several possible reasons for this. Decreases in the efficacy of 1) the adjuvant, 2) antigen presentation, 3) B cell memory development after the primary injection, or 4) overall cell death in B cell counts are all possibilities. The approval of our last flight experiment (where both the primary and secondary inoculations occur in orbit) is crucial to help identify the actual mechanism.

Summary of Work: 2020 - 2021

COVID-19 Related Delays

Loma Linda University went on COVID-related lockdown in March 2020. At that time, all staff were strongly encouraged to work from home if possible. Much of the work associated with this grant involved tissue processing and data analysis. While the data analysis could continue (we were able to get permission for our postdoc to connect remotely to our server), the tissue processing was put on hold. This coincided with a shutdown at the Kansas State University Integrated Genomics Facility for both COVID-related issues and a temporary university-wide power failure that halted processing for several weeks.

Due to the uncertainty related to COVID, both of our postdocs (Drs. Trisha Rettig and John Tan) opted to move on to other positions. Dr. Rettig left for a position in industry on 10/5/2020. Dr. Tan moved on to a faculty position within LLU on 6/30/2020. Similarly, our research technician, Nina Nishiyama, moved on to go to graduate school at the University of North Carolina on July 17, 2020.

Summary of Work 2021 - 2023

Personnel Changes

In 2021, we requested an augmentation to continue the work on this project due to a variety of COVID-related delays and personnel changes. As part of this request, we included salary for half of a technician, a postdoc and a grad student. The funding was approved and the revised cooperative agreement was signed in December of 2021. For well over a year, we were unable to identify a grad student and post-doc to continue working on this project. This was likely due to 1) the difficulties associated with COVID, 2) the national downturn in grad school applications, and 3) the timing for the additional funding (i.e., in the middle of Loma Linda University's academic year). We were finally able to identify both a grad student and post-doc at the end of 2022. However, shortly thereafter, our grad student had to take a prolonged leave of absence due to health-related issues. He returned to work in the summer of 2023.

Changes in Antibody Repertoire Analysis

After several delays and mishaps beyond our control, we had to make a change in our analysis, switching our antibody repertoire analysis away from the facility at Kansas State University and going with the company iRepertoire. For the first 4-5 months of 2022, we dealt with supply chain related delays trying to get the samples checked and prepped to send. We were finally able to send these tissues out to begin processing in the summer of 2022.

Tissue Collection with BSP

For about three weeks in October of 2021, personnel from the Biospecimen Sharing Program (BSP) at NASA Ames Research Center (Ames) traveled to Loma Linda University to help with the tissue collection from the RR-12/TARBIS

mice (frozen in orbit at the time of the mission). This was originally supposed to happen back in 2019 but was delayed considerably for a variety of reasons, including COVID-related facility shutdowns. These tissues were either shipped to Ames and added to the BSP tissue archives or shipped to co-investigators for analysis.

Additional Opportunities in Collaboration with Dr. Xiao Wen “Vivian” Mao

In January-February of 2022, several members of the TARBIS team joined Dr. Xiao Wen Mao in support of her RR-18 Live Animal Return (LAR) tissue collection at the Roskamp Institute in Florida (NASA Grant #NNX15AB41G). Although Dr. Pecaut is a Co-investigator on Dr. Mao’s grant, he also collected additional tissues as part of the BSP (specifically, spleen). Because our RR-12/TARBIS tissues were collected and frozen in orbit, we were not able to perform any functional assays or flow cytometry-based phenotype analysis. Fortunately, Dr. Mao’s flight parameters were very similar to one of our RR-12 flight experiments (i.e., ~ 4 week flight time). Therefore, we used this opportunity to do phenotyping and cytokine expression to complement our antibody analysis from our previous flight. This also complements our ground-based centrifuge study ("Spaceflight Impacts the Microbiome-Brain Axis" or “SIMBA”, NASA Grant #: 80NSSC19K1038) which was also for 4 weeks. This analysis is ongoing.

In April-May of 2022, Dr. Mao’s second RR-18 dissection occurred. This study was designed to look at the recovery responses 2-3 months after landing. Although we do not have RR-12/TARBS data that would directly correspond with this recovery period, we felt knowing the phenotype distribution and cytokine responses would inform us of the long-term consequences of spaceflight on populations critical to the antibody response.

Current Status

Throughout the final months of this project, both our post-doc and graduate student have been working on analyzing the data generated by iRepertoire. This analysis is ongoing even though the funds for this project have ceased.

REFERENCES

Chapes SK, Mastro AM, Sonnenfeld G, Berry WD. Antiorthostatic suspension as a model for the effects of spaceflight on the immune system. Journal of Leukocyte Biology. 1993;54(3):227-35. pmid:8371052.

Bibliography Type:	Description: (Last Updated: 08/04/2025)
Articles in Peer-reviewed Journals	Oyefeso FA, Goldberg G, Opoku N, Vazquez M, Bertucci A, Chen Z, Wang C, Muotri AR, Pecaut MJ. "Effects of acute low-moderate dose ionizing radiation to human brain organoids." PLoS One. 2023 May 31;18(5):e0282958. https://doi.org/10.1371/journal.pone.0282958 ; PMID: 37256873; PMCID: PMC10231836 , May-2023
Articles in Peer-reviewed Journals	Veliz AL, Mamoun L, Hughes L, Vega R, Holmes B, Monteon A, Bray J, Pecaut MJ, Kearns-Jonker M. "Transcriptomic effects on the mouse heart following 30 days on the International Space Station." Biomolecules. 2023 Feb 15;13(2):371. https://doi.org/10.3390/biom13020371 ; PMID: 36830740; PMCID: PMC9953463. , Feb-2023
Articles in Peer-reviewed Journals	Paul AM, Overbey EG, da Silveira WA, Szewczyk N, Nishiyama NC, Pecaut MJ, Anand S, Galazka JM, Mao XW. "Immunological and hematological outcomes following protracted low dose/low dose rate ionizing radiation and simulated microgravity." Sci Rep. 2021 Jun 1;11(1):11452. https://doi.org/10.1038/s41598-021-90439-5 ; PMID: 34075076; PMCID: PMC8169688 , Jun-2021
Articles in Peer-reviewed Journals	Holley JM, Stanbouly S, Pecaut MJ, Willey JS, Delp M & Mao XW. "Characterization of gene expression profiles in the mouse brain after 35 days of spaceflight mission." NPJ Microgravity. 2022 Aug 10;8(1):35. https://doi.org/10.1038/s41526-022-00217-4 ; PMID: 35948598; PMCID: PMC9365836 , Aug-2022
Articles in Peer-reviewed Journals	Veliz AL, Hughes L, Carrillo D, Pecaut MJ, Kearns-Jonker M. "Immunization induces inflammation in the mouse heart during spaceflight." BMC Genomics. 2025 Mar 10;26:229. https://doi.org/10.1186/s12864-025-11426-y ; PMID: 40065216; PMCID: PMC11892206 , Mar-2025